

MCMURDO STATION, Antarctica

WMO: 896640

Lat: 77.850S

Lon: 166.667E

Elev: 24

StdP: 101.04

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-36.4	-34.7	-42.4	0.1	-35.4	-40.6	0.1	-33.6	21.3	-17.3	17.6	-21.0	3.7	80	1.231

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.4	3.0	-0.8	1.8	-1.3	0.9	-1.8	0.0	2.2	-0.9	1.2	-1.6	0.5	2.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-2.3	3.1	0.3	-3.2	2.9	-0.1	-4.2	2.7	-0.7	9.7	2.4	8.2	1.3	7.0	0.7	3.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 16.2	13.1	11.2	DB	-39.2	5.1	3.2	1.8	-41.5	6.4	-43.4	7.5	-45.2	8.5	-47.5	9.8	(4)
(5)			WB	-39.0	1.3	2.4	1.6	-40.7	2.5	-42.1	3.4	-43.5	4.3	-45.2	5.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-16.2	-2.6	-8.4	-16.5	-20.6	-21.1	-23.2	-24.7	-25.0	-23.3	-17.1	-8.6	-2.7		
	DBStd	9.43	2.68	3.92	4.61	4.88	5.89	5.74	5.92	6.37	5.59	4.36	4.19	2.66		
	HDD10.0	9559	392	516	823	917	966	994	1074	1086	1000	841	558	393		
	HDD18.3	12601	651	749	1081	1167	1224	1244	1332	1344	1250	1099	808	651		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	4.8	4.0	5.1	5.4	4.8	4.9	5.2	4.8	4.7	4.9	5.0	4.7	4.2		
(15) Precipitation	PrecAvg	176	16	22	14	16	22	26	15	12	11	14	12	9		
	PrecMax	343	72	52	60	40	78	134	68	47	35	55	42	22		
	PrecMin	66	0	0	0	0	0	0	1	0	1	0	0	0		
	PrecStd	71	18	14	13	14	20	31	15	12	8	12	12	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	1.0	-5.1	-7.9	-6.4	-7.3	-8.0	-8.8	-7.5	-5.3	1.1	4.6		
		MCWB	1.0	-2.4	-7.0	-9.5	-7.7	-8.8	-8.7	-10.2	-9.0	-7.1	-2.4	0.6		
	2%	DB	3.2	-1.0	-7.0	-9.6	-9.2	-10.6	-10.9	-11.2	-10.4	-7.9	-0.5	3.1		
		MCWB	-0.4	-3.5	-8.4	-10.7	-10.4	-11.6	-12.0	-12.3	-11.7	-9.8	-3.5	-0.9		
	5%	DB	1.9	-2.2	-8.6	-11.6	-10.9	-12.3	-13.2	-13.2	-12.5	-9.7	-1.9	2.1		
		MCWB	-1.1	-4.5	-9.9	-12.7	-12.1	-13.3	-14.1	-14.2	-13.6	-11.1	-4.3	-1.3		
	10%	DB	1.0	-3.3	-10.0	-13.6	-12.8	-14.3	-15.8	-15.0	-14.8	-11.1	-3.1	1.1		
		MCWB	-1.6	-5.4	-11.2	-14.5	-13.8	-15.3	-16.6	-15.7	-15.8	-12.3	-5.3	-1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	-1.7	-6.7	-9.4	-7.5	-9.1	-8.6	-9.9	-8.8	-7.4	-2.2	1.1		
		MCDB	4.5	-0.1	-5.2	-8.5	-6.3	-7.8	-7.7	-8.9	-8.7	-5.7	-0.3	4.2		
	2%	WB	0.3	-3.3	-8.4	-10.6	-10.1	-11.5	-12.4	-12.1	-12.2	-10.0	-3.2	-0.3		
		MCDB	2.8	-1.4	-7.1	-9.5	-9.1	-10.8	-11.5	-11.2	-11.1	-8.5	-1.2	2.1		
	5%	WB	-0.5	-4.4	-9.9	-12.7	-12.0	-13.4	-14.7	-14.1	-13.9	-11.3	-4.4	-1.0		
		MCDB	1.6	-2.5	-8.7	-11.6	-11.0	-12.4	-13.7	-13.3	-12.9	-10.1	-2.4	1.3		
	10%	WB	-1.2	-5.5	-11.2	-14.5	-13.7	-15.4	-16.9	-15.7	-15.9	-12.5	-5.5	-1.6		
		MCDB	0.6	-3.6	-10.0	-13.6	-12.7	-14.5	-16.2	-14.9	-15.0	-11.4	-3.3	0.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	4.6	4.8	5.0	5.3	5.4	5.5	5.8	6.2	5.7	5.1	4.1		
		MCDBR	5.3	6.0	5.3	7.1	6.4	7.1	7.2	6.7	8.1	6.6	5.2	5.3		
	5% WB	MCWBR	3.8	4.8	4.7	6.6	5.8	6.5	6.9	6.2	7.4	5.9	4.3	3.6		
		MCDBR	4.9	5.7	5.2	7.1	6.0	7.1	7.6	6.5	7.7	6.6	5.3	4.8		
		MCWBR	3.8	4.8	4.7	6.7	5.6	6.6	7.1	6.2	7.2	6.0	4.4	3.6		
(40) Clear-Sky Solar Irradiance	taub	0.236	0.218	0.174	0.113	N/A	N/A	N/A	N/A	0.111	0.165	0.196	0.217	0.234		
	taud	2.248	2.262	2.177	1.896	N/A	N/A	N/A	N/A	1.877	2.126	2.235	2.256	2.246		
	Ebn at Noon	964	897	765	112	0	0	0	0	96	762	928	988	990		
	Edh at Noon	98	76	46	3	0	0	0	0	3	46	77	96	104		
(44) All-Sky Solar Radiation	RadAvg	8.00	4.61	1.53	0.13	0.00	0.00	0.00	0.00	0.02	0.86	3.56	7.36	9.31		
	RadStd	0.22	0.18	0.07	0.01	0.00	0.00	0.00	0.00	0.00	0.05	0.13	0.22	0.27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page